

$$\begin{array}{ccc}
\mathcal{D}(F(\iota F X \otimes_c Y), B) & \xrightarrow{\mathcal{D}(F(\eta_X \otimes_c Y), B)} & \mathcal{D}(F(X \otimes_c Y), B) \\
\cong \downarrow & & \downarrow \cong \\
\mathcal{C}(\iota F X \otimes_c Y, \iota B) & \xrightarrow{\mathcal{C}(\eta_X \otimes_c Y, \iota B)} & \mathcal{C}(X \otimes_c Y, \iota B) \\
\cong \downarrow & & \downarrow \cong \\
\mathcal{C}(\iota F X, [Y, \iota B]_c) & \xrightarrow{\mathcal{C}(\eta_X, [Y, \iota B]_c)} & \mathcal{C}(X, [Y, \iota B]_c) \\
\mathcal{C}(\iota F X, \eta_{[Y, \iota B]_c}) \downarrow \cong & & \cong \downarrow \mathcal{C}(X, \eta_{[Y, \iota B]_c}) \\
\mathcal{C}(\iota F X, \iota F[Y, \iota B]_c) & \xrightarrow{\mathcal{C}(\eta_X, \iota F[Y, \iota B]_c)} & \mathcal{C}(X, \iota F[Y, \iota B]_c).
\end{array}$$